SISTEM INFORMASI AKADEMIK POLINEMA

PROJECT MANAGEMENT PLAN

POLITEKNIK NEGERI MALANG
Jl. Soekarno Hatta no. 9, Malang, 65141
polinema.ac.id

VERSION 1.0.0

19/11/2024

VERSION HISTORY

VERSION	APPROVED BY	REVISION DATE	DESCRIPTION OF CHANGE	AUTHOR
1.0.0	Project Sponsor	Nov 19, 2024	Initial Version	Project Manager

PREPARED BY	TITLE	DATE
Pramana Yoga Saputra		19/11/2024

APPROVED BY	TITLE	DATE

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1. EXECUTIVE SUMMARY

The "Academic Information System for Politeknik Negeri Malang" project aims to create a web-based system for managing academic activities, including course management, student enrollment, and administrative tasks. Built using the Laravel Framework, this project will streamline academic workflows, ensure data integrity, and enhance accessibility for users. The project is budgeted at Rp 50,000,000 and is scheduled to be completed in three months.

2. PROJECT MANAGEMENT APPROACH AND GOVERNANCE

The project will adopt an agile development methodology with iterative development cycles. A project manager will oversee milestones and resource allocation, while the team follows a predefined schedule and governance processes. Vendors will supply hosting and domain services, with Trello used for project tracking.

Roles and Responsibilities:

- **Project Manager**: Overall oversight, resource allocation, and progress tracking.
- **UI/UX Designer**: Wireframing and prototyping.
- **Frontend Developer**: UI implementation using HTML, CSS, and Bootstrap.
- **Backend Developer**: API and database development using Laravel.
- **QA Tester**: Testing and quality assurance.

Assumptions and Constraints:

- Free-tier tools will suffice for development and testing.
- The budget and timeline are fixed.

2.1 PROJECT SCOPE

The scope includes:

- Development of a web-based academic system.
- Key functionalities: student registration, course management, reporting, and administrative tools.
- User-friendly interface compatible across devices.

Out of scope: Mobile application development and features requiring external integrations beyond budget.

2.2 DELIVERABLES

- Functional Academic Information System accessible via web.
- Documentation for users and administrators.
- Deployment and training for system use.

2.3 WORK BREAKDOWN STRUCTURE (WBS)

WBS Code	Phase/Task	Duration	Start Date	End Date
1.1	**Project Initiation**			
1.1.1	Requirement Gathering	1 week	Sep 2, 2024	Sep 8, 2024
1.1.2	Define Scope & Features	1 week	Sep 9, 2024	Sep 15, 2024
1.1.3	Project Plan & Schedule	1 week	Sep 16, 2024	Sep 22, 2024
1.2	**Design Phase**			
1.2.1	Wireframing & Mockups	2 weeks	Sep 23, 2024	Oct 6, 2024
1.2.2	Finalizing UI/UX Design	1 week	Oct 7, 2024	Oct 13, 2024
1.3	**Development Phase**			
1.3.1	**Frontend Development**			
1.3.1.1	Base Layouts & Components	2 weeks	Oct 14, 2024	Oct 27, 2024
1.3.1.2	Implement UI Designs	2 weeks	Oct 28, 2024	Nov 10, 2024
1.3.1.3	Responsive Testing	1 week	Nov 11, 2024	Nov 17, 2024
1.3.2	**Backend Development**			
1.3.2.1	Database Design & Setup	1 week	Oct 14, 2024	Oct 20, 2024
1.3.2.2	API Development	2 weeks	Oct 21, 2024	Nov 3, 2024
1.3.2.3	Authentication	1 week	Nov 4, 2024	Nov 10, 2024
1.4	**Integration & Testing**			
1.4.1	Frontend-Backend Integration	1 week	Nov 11, 2024	Nov 17, 2024
1.4.2	Functionality Testing	1 week	Nov 18, 2024	Nov 24, 2024
1.4.3	Bug Fixes & Optimization	2 weeks	Nov 11, 2024	Nov 24, 2024
1.5	**Deployment**			
1.5.1	Setup Hosting & Domain	1 week	Nov 11, 2024	Nov 17, 2024
1.5.2	Final Deployment	1 week	Nov 18, 2024	Nov 24, 2024
1.5.3	User Training & Docs	1 week	Nov 18, 2024	Nov 24, 2024

2.4 STAKEHOLDER ANALYSIS

For the development project of the Academic Information System (SIAKAD) at POLINEMA (Politeknik Negeri Malang), here are the key stakeholders involved:

1. Primary Stakeholders

1.1. Management

- Director of POLINEMA
  The main decision-maker, ensuring the project aligns with the institution's vision and mission. Responsible for budget approval and providing strategic direction for the system.
- Vice Directors (Academic, Financial, and Student Affairs)
  Provide input regarding system requirements based on academic policies, financial management, and student services.

1.2. End-Users

- Lecturers
  Use SIAKAD for entering grades, managing class schedules, and monitoring student progress. Require a user-friendly interface and accurate data.
- Students
  Use the system for course registration, viewing schedules, academic status, and fee payments. Expect a fast, intuitive, and mobile-friendly system.
- Administrative Staff
  Handle academic administration tasks (e.g., scheduling, managing student data, and transcripts). Need features to enhance daily operational efficiency.

2. Technical Team

- Development Team (Internal or External)
  Responsible for designing, developing, and implementing the system. Includes roles such as project manager, developers, UI/UX designers, and QA testers.
- POLINEMA IT Team
  Manages servers, networks, and ensures system security. Plays a role in integrating SIAKAD with other services, such as the student portal or LMS.

3. Supporting Parties

- Finance Department
  Oversees the payment module, integration with banks, and financial reporting.
- Academic Affairs Department
  Contributes to defining specific requirements, such as the academic calendar, curriculum, and scheduling modules.
- Vendors or Service Providers
  External providers for services such as hosting, domain registration, or APIs (e.g., integration with library or financial systems).

4. External Stakeholders

- Ministry of Education, Culture, Research, and Technology (Kemendikbudristek)
  Plays a role in regulation, academic data reporting (e.g., PDDikti), and setting standards for information system management.
- Partner Banks
  Providers of payment services (integration of payment gateways for tuition fees and other administrative payments).
- Potential Users
  Alumni and prospective students who might use certain features (e.g., registration, transcripts).

5. System Evaluation and Testing

- System Evaluation Team
  A mix of stakeholders above to conduct trials and provide feedback before the system is officially launched.

2.5 SCHEDULE BASELINE

The **SIAKAD project** is planned to follow the approved schedule baseline as follows:

1. The **Project Initiation** phase will begin with *Requirement Gathering* from **September 2 to September 8, 2024** (1 week). This will be followed by defining the *Scope and Features* from **September 9 to September 15, 2024** (1 week) and preparing the *Project Plan and Schedule* from **September 16 to September 22, 2024** (1 week).
2. During the **Design Phase**, the team will work on *Wireframing and Mockups* from **September 23 to October 6, 2024** (2 weeks). Once complete, the *Final UI/UX Design* will be finalized from **October 7 to October 13, 2024** (1 week).
3. The **Development Phase** consists of two parallel streams: frontend and backend development.
    - For **Frontend Development**, the team will work on *Base Layouts and Components* from **October 14 to October 27, 2024** (2 weeks), followed by *UI Implementation* from **October 28 to November 10, 2024** (2 weeks). Finally, *Responsive Testing* will take place from **November 11 to November 17, 2024** (1 week).
    - For **Backend Development**, *Database Design and Setup* will be completed from **October 14 to October 20, 2024** (1 week), followed by *API Development* from **October 21 to November 3, 2024** (2 weeks) and *Authentication Setup* from **November 4 to November 10, 2024** (1 week).
4. The **Integration and Testing Phase** will involve *Frontend-Backend Integration* from **November 11 to November 17, 2024** (1 week), followed by *Functionality Testing* from **November 18 to November 24, 2024** (1 week). Simultaneously,

Bug Fixes and Optimization will run from **November 11 to November 24, 2024** (2 weeks).

5. The final **Deployment Phase** includes setting up *Hosting and Domain* from **November 11 to November 17, 2024** (1 week), conducting the *Final Deployment* from **November 18 to November 24, 2024** (1 week), and completing *User Training and Documentation* during the same timeframe from **November 18 to November 24, 2024** (1 week).

2.6 MILESTONE LIST

MILESTONE	DESCRIPTION	DATE
Project Plan Completion	Finalization of project plan document	Sep 22, 2024
UI/UX Design Completion	Delivery of wireframes and prototypes	Oct 13, 2024
Frontend Implementation	Completion of base layouts and components	Nov 10, 2024
Full Integration	Frontend and backend integration complete	Nov 17, 2024
Final Deployment	System ready for use	Nov 24, 2024

2.7 CHANGE MANAGEMENT PLAN

1. Change Control Process

1. **Submission of Change Requests**
    - **Who Can Submit Changes:**
      Changes can be requested by project stakeholders, including the project manager, team members (e.g., developers, designers), and external stakeholders (e.g., clients or end-users).
    - Change requests must be documented using the *Change Request Form (CRF)*, which includes:
        - A description of the proposed change.
        - The reason for the change.
        - The impact on project scope, timeline, cost, and resources.
2. **Evaluation of Change Requests**
    - Change requests will be evaluated by the **Change Control Board (CCB)** or designated decision-makers. The evaluation includes:
        - Feasibility analysis.
        - Impact assessment on project scope, timeline, budget, and quality.
        - Resource availability check.

- The project manager facilitates the evaluation process by providing input from the project team and stakeholders.

3. **Approval Process**
   - All major changes must be approved by the **Change Control Board (CCB)**.
     - For minor changes (e.g., UI modifications or small bug fixes), the project manager has the authority to approve or reject.
   - The CCB will include:
     - Project Manager (Chair).
     - Key Team Leads (e.g., Frontend and Backend Developers).
     - Stakeholder Representatives.
4. **Implementation of Approved Changes**
   - Approved changes will be added to the project plan, with updates to the schedule baseline, budget, and other relevant documents.
   - The project manager will assign tasks for implementation and monitor progress.
5. **Communication of Changes**
   - Approved changes will be communicated to all relevant stakeholders via:
     - Project management tools (e.g., Trello or Slack).
     - Email notifications.
     - Regular project status meetings.
   - Updates to project documentation will be shared in the central repository.
6. **Tracking Changes**
   - Changes will be tracked using a *Change Log* that records:
     - Change request ID.
     - Date of request and implementation.
     - Description of the change.
     - Approval status.
     - Implementation status.

2. Roles and Responsibilities

Role	Responsibility
Change Requester	Submits a detailed change request form.
Project Manager	Facilitates the evaluation process, approves minor changes, and monitors execution.
Change Control Board (CCB)	Evaluates and approves/rejects major changes.
Development Team	Implements approved changes and provides impact assessments for proposed changes.

3. Tools and Documentation

- **Change Request Form (CRF):** Standardized template for submitting requests.
- **Change Log:** A centralized record of all submitted, approved, and implemented changes.
- **Project Management Tools:** Trello and Slack for tracking and communication.

2.8 PROJECT SCOPE MANAGEMENT PLAN

Scope Management Approach

The scope management process for the SIAKAD project will involve the following activities:

1. **Scope Planning**
   The project scope will be defined during the initiation phase, based on requirements gathered from stakeholders, including Politeknik Negeri Malang. The deliverables, boundaries, and exclusions will be clearly documented in the Project Scope Statement.
2. **Scope Definition**
   Detailed scope definition will be achieved through consultations with stakeholders to create:
   - A detailed **Work Breakdown Structure (WBS)**.
   - A comprehensive **Requirements Document** to outline system functionalities, including user authentication, course registration, grade submission, and academic reporting.
3. **Scope Validation**
   Stakeholders, including representatives from Politeknik Negeri Malang, will review and formally approve the project deliverables to ensure alignment with agreed requirements. Validation will occur at key milestones:
   - Completion of the design phase (UI/UX mockups).

  - After the development phase (frontend and backend implementation).
  - Upon final deployment of the system.
4. **Scope Control**
   The project manager will monitor and manage changes to the scope using a formal **Change Control Process**. Changes will only be implemented after approval by the **Change Control Board (CCB)** to minimize disruption.

Scope Statement

The SIAKAD project will deliver a **web-based Academic Information System** for Politeknik Negeri Malang, developed using the Laravel framework. The system will provide functionalities for students, faculty, and administrators, including:

- **Students**:
  - View personal information and academic performance.
  - Register for courses and submit assignments.
- **Faculty**:
  - Input grades and attendance.
  - Access class lists and reports.
- **Administrators**:
  - Manage user accounts, course offerings, and academic records.
  - Generate institutional reports.

Key Deliverables Include:

1. Functional Academic Information System (SIAKAD).
2. Responsive user interface aligned with approved UI/UX designs.
3. Database setup and integration for secure data management.
4. Hosting setup and final deployment.
5. Training materials and user documentation.

Roles and Responsibilities

Role	**Responsibility**
Project Manager	Oversees scope definition, validation, and control. Monitors for scope creep.
UI/UX Designer	Develops wireframes and mockups to meet stakeholder requirements.

Frontend & Backend Developers	Implements system functionality within defined scope.
QA Tester	Validates that deliverables meet stakeholder expectations and requirements.
Stakeholders	Provides feedback, approves deliverables, and requests changes when necessary.

Scope Control Process

The scope control process includes:

1. **Change Requests**: All scope changes must be submitted via a *Change Request Form (CRF)*.
2. **Impact Analysis**: Changes will be evaluated based on their impact on the project's schedule, cost, and quality.
3. **Approval Process**:
   - Minor changes may be approved by the Project Manager.
   - Major changes require CCB approval.
4. **Documentation**: Approved changes will be documented in the updated Project Scope Statement and shared with all team members.

Work Breakdown Structure (WBS)

The WBS defines all project deliverables and associated tasks. It will guide project execution and tracking. Key phases include:

1. **Initiation Phase**: Requirement gathering, scope definition.
2. **Design Phase**: UI/UX development and wireframing.
3. **Development Phase**: Frontend and backend implementation.
4. **Testing and Validation**: Functional and system testing.
5. **Deployment**: Final deployment and user training.

Scope Management Metrics

- **Scope Creep**: Any deviations from the approved scope will be tracked and evaluated.
- **Stakeholder Satisfaction**: Measured via milestone reviews and final deliverable approval.
- **Deliverable Completion**: Each deliverable will be marked complete only after formal stakeholder validation.

Approval Process

The Project Scope Statement and all subsequent changes to the scope will require approval from:

- **Primary Stakeholders**: Representatives from Politeknik Negeri Malang.
- **Project Manager**: Ensures feasibility and alignment with project objectives.

3. COMMUNICATION MANAGEMENT PLAN

Contact Information

NAME	TITLE	EMAIL	OFFICE PHONE
Project Manager	Responsible for overseeing the project progress.	pm@polinema.ac.id	(0341) 123456
UI/UX Designer	Handles the design aspects of the project.	design@polinema.ac.id	(0341) 123457
Frontend Developer	Implements the user interface and frontend logic.	frontend@polinema.ac.id	(0341) 123458
Backend Developer	Develops APIs and server-side logic.	backend@polinema.ac.id	(0341) 123459
QA Tester	Ensures system quality and functionality.	qa@polinema.ac.id	(0341) 123460
Project Manager	Responsible for overseeing the project progress.	pm@polinema.ac.id	(0341) 123456
Academic Administrator	Manages academic data and administrative operations.	admin.academic@polinema.ac.id	(0341) 123461
Head of Department	Oversees departmental activities and provides project insights.	hod@polinema.ac.id	(0341) 123462
Academic Supervisor	Advises students and provides academic mentorship.	advisor@polinema.ac.id	(0341) 123463

Communication Details

COMMUNICATION TYPE	DESCRIPTION	FREQUENCY	MESSAGE DISTRIBUTION	DELIVERABLE	DELIVERABLE OWNER
Kickoff Meeting	Introduce the team and align on project objectives.	Once (Project Start)	In-person and virtual (Google Meet)	Meeting Minutes	Project Manager
Weekly Team Meetings	Review progress, discuss challenges, and assign tasks for the week.	Weekly	Virtual (Google Meet)	Progress Updates, Action Items	Project Manager
Stakeholder Updates	Provide key stakeholders with project status and milestones.	Bi-weekly	Email and shared documents	Status Reports	Project Manager

Design Reviews	Present and validate UI/UX designs with stakeholders.	Once per phase	In-person/Virtual	Approved UI/UX Designs	UI/UX Designer
Development Updates	Summarize development progress and align on upcoming work.	Bi-weekly	Email/Slack	Code Checkpoints	Frontend & Backend Developers
Testing Reports	Share findings from functionality and performance testing.	Per testing phase	Email and shared documents	Bug Reports, Testing Results	QA Tester
Final Review Meeting	Review project deliverables with stakeholders and finalize for deployment.	End of project	In-person/Virtual	Final Approval Document	Project Manager

Message Distribution Methods

- Google Meet: For virtual meetings and progress discussions.
- Email: For official communication, including meeting summaries, updates, and status reports.
- Slack: For instant team communication and task tracking.
- Google Drive: For sharing deliverables and important project documents.

Escalation Process

In case of delays, unresolved issues, or urgent matters:

1. The issue is reported to the Project Manager via email or Slack.
2. The Project Manager evaluates the severity and decides whether to escalate to stakeholders.
3. Escalation meetings will be scheduled as required to resolve the matter.

4. RESOURCE MANAGEMENT PLAN

1. Resource Overview

The SIAKAD project will utilize both human and material resources to ensure timely and efficient delivery. Human resources include the project team and external contributors, while material resources include software tools, hardware, and hosting infrastructure.

2. Human Resource Management

Role	**Responsibilities**	**Time Commitment**	**Cost**
Project Manager	Define project scope, manage team coordination, track progress, and control budget.	Part-time (~10 hrs/week)	Rp 9,000,000 (3 months)
UI/UX Designer	Design wireframes, prototypes, and user-friendly interfaces using tools like Figma.	Full-time (4–6 weeks)	Rp 4,000,000 (one-time)
Frontend Developer	Develop the user interface, ensure responsiveness, and implement design using HTML, CSS, and JavaScript.	Full-time (3 months)	Rp 15,000,000 (3 months)
Backend Developer	Set up the database, develop APIs, handle authentication, and server-side logic.	Full-time (3 months)	Rp 15,000,000 (3 months)
QA Tester	Perform functionality, usability, and security testing, and report bugs.	Part-time (~10 hrs/week)	Rp 2,000,000 (one-time)

3. Material Resource Management

Resource Type	**Description**	**Cost**
Wireframing Tools	Figma (free version) for prototyping and designing user interfaces.	Free
Project Management Tools	Trello (free version) for task and project tracking.	Free
Frontend Development Tools	HTML, CSS, JavaScript, Bootstrap, Google Fonts, FontAwesome.	Free
Backend Development Tools	Laravel (PHP), MySQL/PostgreSQL, Auth0 (free tier).	Free
Hosting and Domain	Niagahoster for hosting and domain registration.	Rp 575,000 (3 months)

Performance Tools	Lighthouse (performance testing) and Sentry (error tracking free tier).	Free
Security Tools	Let's Encrypt (SSL certificate) and OWASP ZAP (free security testing tool).	Free
Email Notifications	SendGrid (free tier for 100 emails/day).	Free
Analytics Tools	Google Analytics (free version).	Free

4. Resource Allocation

Key Allocation Details:

- The Project Manager will oversee all resources, ensuring balanced workload and efficient use.
- The UI/UX Designer will collaborate with the developers during the first design phase to finalize interfaces before the development phase begins.
- Frontend and Backend Developers will work simultaneously during development but collaborate closely during integration and testing.
- The QA Tester will be involved in the final month for testing and reporting bugs.
- Hosting and domain registration will be completed during the deployment phase, ensuring readiness for the final release.

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5. Conflict Resolution

Any conflicts related to resource availability or allocation will be addressed by the Project Manager. Escalations will be resolved during weekly team meetings or directly with stakeholders if required.

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6. Resource Utilization Monitoring

- Progress will be tracked weekly using Trello, ensuring tasks and milestones are completed as planned.
- Budget adherence will be monitored monthly to ensure resources are used efficiently and within the allocated budget of Rp 50,000,000.
- Adjustments to resource plans will be discussed and approved in team meetings if necessary.

5. HUMAN RESOURCES MANAGEMENT PLAN

1. Recruitment Plan

- **Internal Team**: Academic Administrator, Head of Department, and Academic Supervisor will be sourced internally.
- **External Hiring**: Freelance designers, developers, and QA testers will be hired for the required roles.

2. Performance Management

- **Key Metrics**: Task completion on time, quality of deliverables, and teamwork.
- **Review Process**: Weekly team meetings and monthly evaluations by the Project Manager.

3. Conflict Resolution

- **Step 1**: The Project Manager discusses the issue with involved parties.
- **Step 2**: Escalate unresolved issues to a team discussion for resolution.
- **Step 3**: Critical conflicts are escalated to stakeholders, if necessary.

4. Resource Allocation Timeline

Role	Project Phase	Timeline
Project Manager	All phases	Sep 2, 2024 – Nov 24, 2024
UI/UX Designer	Design phase	Sep 23, 2024 – Oct 13, 2024
Frontend Developer	Development, integration, and testing phases	Oct 14, 2024 – Nov 24, 2024
Backend Developer	Development, integration, and testing phases	Oct 14, 2024 – Nov 24, 2024
QA Tester	Testing phase	Nov 11, 2024 – Nov 24, 2024

5.1 PROJECT STAFF LIST

NAME	TITLE	EMAIL	PHONE
Aditya Putra	Project Manager	pm@polinema.ac.id	(0341) 123456
Nina Kartika	UI/UX Designer	design@polinema.ac.id	(0341) 123457
Rizky Mahendra	Frontend Developer	frontend@polinema.ac.id	(0341) 123458
Budi Santoso	Backend Developer	backend@polinema.ac.id	(0341) 123459
Ayu Pratiwi	QA Tester	qa@polinema.ac.id	(0341) 123460

5.2 RESOURCE REQUIREMENT CALENDAR

Resource	Phase	Required From	Required Until	Comments
Project Manager	All phases	Sep 2, 2024	Nov 24, 2024	Part-time, manages overall project progress and coordination.
UI/UX Designer	Design phase	Sep 23, 2024	Oct 13, 2024	Full-time during wireframing and UI/UX design tasks.
Frontend Developer	Development & Testing	Oct 14, 2024	Nov 24, 2024	Full-time, responsible for implementing the frontend.
Backend Developer	Development & Testing	Oct 14, 2024	Nov 24, 2024	Full-time, handles backend development and integration.
QA Tester	Testing phase	Nov 11, 2024	Nov 24, 2024	Part-time, conducts testing for functionality and security.
Hosting Service	Deployment phase	Nov 11, 2024	Nov 24, 2024	Hosting platform (Niagahoster) for deployment.
Domain Name	Deployment phase	Nov 11, 2024	Nov 24, 2024	Domain registration for system accessibility.
Figma (Wireframing Tool)	Design phase	Sep 23, 2024	Oct 13, 2024	Free plan used for wireframing and UI/UX design.
Laravel Framework	Development & Testing	Oct 14, 2024	Nov 24, 2024	Framework used for backend development.
MySQL Database	Development & Testing	Oct 14, 2024	Nov 24, 2024	Database for academic record storage.
Trello (Task Management)	All phases	Sep 2, 2024	Nov 24, 2024	Free version used for task and project management.
GitHub (Version Control)	Development & Deployment	Oct 14, 2024	Nov 24, 2024	Free plan for version control and collaboration.
Lighthouse	Testing phase	Nov 11, 2024	Nov 24, 2024	Used for performance testing.
SendGrid	Deployment phase	Nov 11, 2024	Nov 24, 2024	Free tier for email notifications (100 emails/day).
Google Analytics	Deployment phase	Nov 18, 2024	Nov 24, 2024	Used for monitoring system usage.

7. SCHEDULE MANAGEMENT PLAN

1. Schedule Development

Methods for Schedule Development:

- The project schedule will be developed using the **Work Breakdown Structure (WBS)** to define tasks, phases, and deliverables.
- Each task will include details such as:
  - Start and end dates.
  - Assigned resources.
  - Dependencies with other tasks.
  - Duration and milestones.
- Dependencies between tasks will be identified (e.g., sequential or parallel tasks), and the critical path will be determined to highlight tasks that directly impact the project completion date.

Key Inputs for Developing the Schedule:

- **Scope Statement:** Provides an overview of the project deliverables and boundaries.
- **Resource Calendar:** Indicates the availability of both human and non-human resources.
- **Effort Estimates:** Duration estimates for each task based on team inputs and historical data.

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2. Schedule Monitoring and Control

Monitoring the Schedule:

- Regular progress updates will be conducted weekly to ensure tasks are on track.
- Any deviations from the baseline will be flagged and analyzed.

Control Processes:

- If a delay occurs, the project manager will analyze its impact on the critical path and propose adjustments.
- Approved changes will result in an updated schedule baseline to maintain accuracy.
- Regular stakeholder meetings will ensure that updates and changes are communicated effectively.

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3. Tools for Schedule Management

The following tools will be used to record, monitor, and update the schedule:

Tool	Purpose	Notes
Trello	Task management and tracking.	Free version will be used for managing task assignments.

Gantt Chart (Excel)	Visualizing the project timeline.	Tracks start/end dates, dependencies, and milestones.
Google Calendar	Scheduling meetings and milestones.	Syncs team activities and reminders.
Slack	Communication and quick updates.	Used for daily coordination and schedule discussions.
Project Reports	Weekly progress reporting.	Shared with stakeholders to track the project's progress.

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4. Schedule Baseline

- The approved project schedule will serve as the **schedule baseline**.
- The schedule baseline includes key milestones, such as:
    - Completion of wireframes and mockups.
    - Completion of frontend and backend development.
    - System integration and testing.
    - Final deployment and user training.
- Any changes to the baseline must be approved through the **Change Management Process**.

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5. Change Management

- Changes to the schedule will be managed through a formal process:
    - Submit a **Change Request Form (CRF)** with details on the proposed schedule change.
    - Analyze the impact on scope, cost, and resources.
    - Secure approval from the project manager and stakeholders.
- Changes will be communicated to the team via Trello and email updates.

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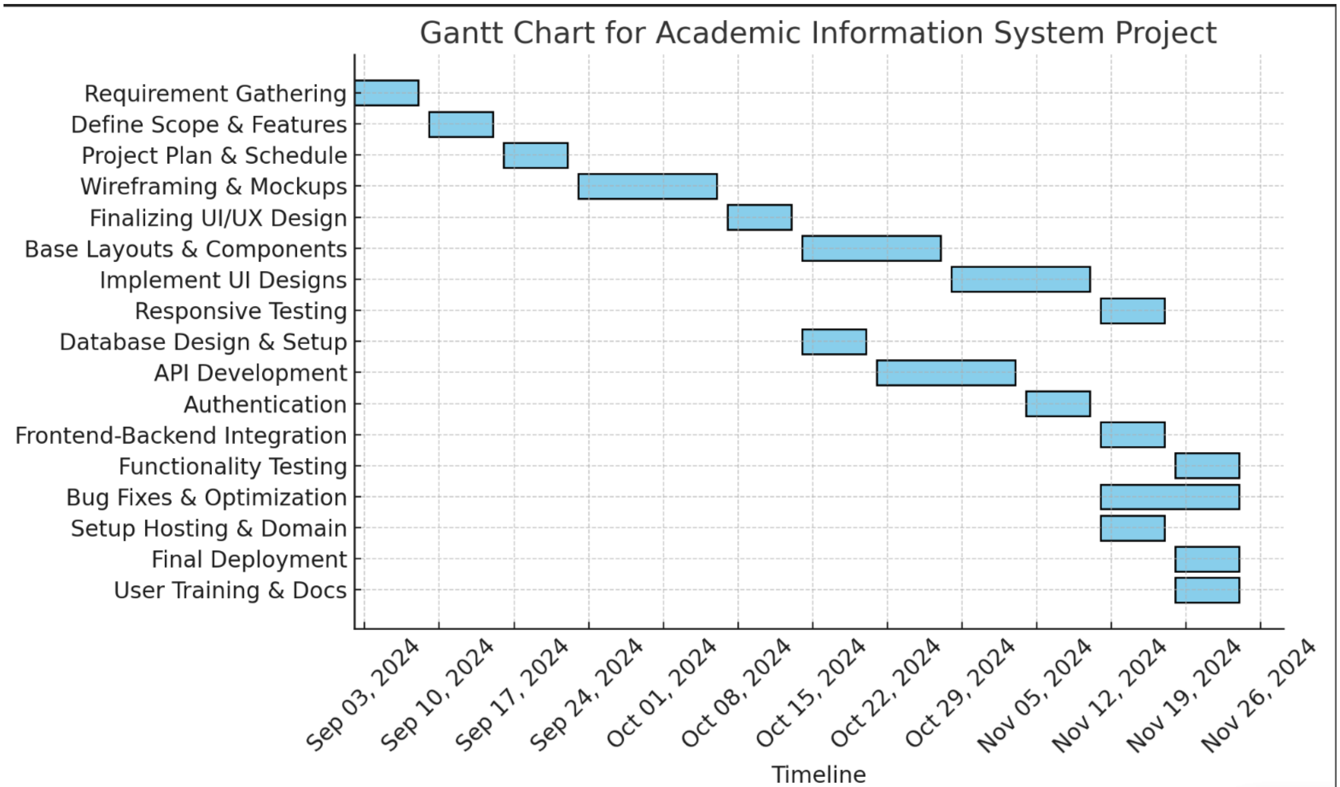
6. Reporting Schedule Progress

- Progress will be reviewed weekly during team meetings.
- A status report will be shared with stakeholders bi-weekly, summarizing:
    - Completed tasks.
    - Delayed or at-risk tasks.
    - Recommendations to address delays.

- Key milestones will trigger validation and sign-off from stakeholders.

7. Gantt Chart



8. QUALITY MANAGEMENT PLAN

1. Quality Objectives

1. Deliver a functional and user-friendly web-based academic information system.
2. Ensure system responsiveness across devices (mobile, tablet, desktop).
3. Maintain data integrity and secure user authentication.
4. Minimize bugs and errors through comprehensive testing.
5. Deliver clear and comprehensive documentation for users and administrators.

2. Quality Standards

The project must adhere to the following quality benchmarks:

- **Performance:** The system should load within 2 seconds under typical academic use.
- **Usability:** The interface should be intuitive for students, faculty, and administrators.

- **Security:** Must comply with industry security standards, including SSL/TLS and secure authentication.
- **Functionality:** All modules (e.g., course registration, grade management) should function as specified.
- **Accessibility:** The system should be accessible across modern browsers and devices.

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3. Quality Control Process

Testing Phases:

1. **Unit Testing**:
   Conducted during development to test individual modules (e.g., authentication, course registration).
2. **Integration Testing**:
   Test how the frontend interacts with backend APIs and the database.
3. **Performance Testing**:
   Use tools like Lighthouse to evaluate system performance under normal and peak loads.
4. **Security Testing**:
   Conduct penetration testing using OWASP ZAP to identify vulnerabilities.
5. **User Acceptance Testing (UAT)**:
   Engage end-users (students, faculty, administrators) to validate that the system meets their needs.

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4. Tools for Quality Management

The following tools will ensure effective quality control and management:

Tool	Purpose
Lighthouse	Performance testing for system load times.
OWASP ZAP	Security testing for vulnerabilities.
Trello	Tracking and managing bug reports.
Google Forms	Collecting user feedback during UAT.
Manual Testing	Validation of functionality and usability.

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5. Roles and Responsibilities

Role	Responsibilities
QA Tester	Conducts testing, prepares test reports, and ensures compliance with quality standards.
Frontend Developer	Fixes bugs identified during testing and ensures responsive design standards.
Backend Developer	Ensures server-side logic and APIs function correctly and securely.
Project Manager	Monitors quality activities and ensures all deliverables meet quality objectives.

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6. Quality Assurance Activities

Documentation Review:

- Validate the requirements document and designs to ensure clarity and feasibility.

Code Reviews:

- Conduct peer reviews of code during the development phase to ensure adherence to best practices.

Testing Reports:

- Prepare reports summarizing test results, including bugs, resolutions, and recommendations.

Regular Audits:

- Perform quality audits during key milestones, such as after design completion and before deployment.

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7. Metrics for Measuring Quality

Metric	Target
Bug-Free Rate	95% of critical bugs resolved before deployment.
Load Time	System loads within 2 seconds on average.
UAT Satisfaction	At least 90% positive feedback during UAT.
Security Compliance	100% of critical vulnerabilities resolved.

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8. Continuous Improvement

Feedback from UAT and post-deployment will be used to identify areas for improvement. Regular updates will incorporate this feedback to ensure the system evolves to meet stakeholder expectations.

9. RISK MANAGEMENT PLAN

1. Risk Management Approach

Risk Identification:

- Conduct brainstorming sessions with project stakeholders and team members to identify potential risks.
- Review project documents (e.g., scope statement, WBS) to anticipate risks in each phase.
- Consider categories such as technical, schedule, cost, resource, and external risks.

Risk Analysis:

- Use qualitative analysis to assess the likelihood and impact of identified risks:
    - **Likelihood:** High, Medium, Low.
    - **Impact:** High, Medium, Low.
- Prioritize risks based on a risk matrix to focus on high-likelihood and high-impact risks.

Risk Tracking:

- Use a **Risk Log** to document and monitor risks.
- Update the Risk Log regularly during weekly team meetings to reflect changes or new risks.

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2. Identified Risk

Risk Category	Potential Risks	Likelihood	Impact
Technical Risks	Bugs or performance issues during development.	Medium	High
	Integration failures between frontend and backend.	Medium	High
Schedule Risks	Delays in UI/UX design or development phases.	High	High
	Limited availability of key team members (e.g., Academic Administrator).	Medium	Medium

Cost Risks	Budget overruns due to additional hosting or tool requirements.	Low	Medium
Resource Risks	Unavailability of skilled resources (e.g., developers, QA testers).	Medium	High
External Risks	Changes in academic requirements or policies by Politeknik Negeri Malang.	Medium	High

3. Risk Mitigation Strategies

Risk	**Mitigation Plan**
Bugs or performance issues	Conduct regular unit and integration testing during development.
Integration failures	Schedule dedicated integration testing phase and involve both frontend and backend teams.
Delays in development	Use a detailed WBS and monitor progress weekly to identify and address delays early.
Limited availability of team members	Assign backup team members or redistribute workload as needed.
Budget overruns	Monitor budget monthly and review expenditures regularly to prevent overspending.
Changes in academic requirements	Maintain clear communication with stakeholders and ensure flexibility to accommodate minor changes.

4. Risk Tracking

Risk Log:
The Risk Log will document all identified risks, including the following details:

1. **Risk ID:** A unique identifier for each risk.
2. **Description:** A clear explanation of the risk.
3. **Likelihood:** High, Medium, or Low.
4. **Impact:** High, Medium, or Low.
5. **Mitigation Plan:** Steps to reduce or eliminate the risk.
6. **Status:** Open, In Progress, or Resolved.

Tracking Tools:

- Trello: Use cards to track risks and assign them to team members for monitoring.

- Weekly Reports: Include risk updates in weekly progress reports to stakeholders.

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5. Contingency Plans

Technical Risks:

- Backup system architecture to quickly address performance bottlenecks.
- Have an external consultant available for high-priority technical issues.

Schedule Risks:

- Add buffer time for critical tasks in the schedule baseline.
- Reallocate resources to ensure tasks are completed on time.

Resource Risks:

- Maintain a list of alternative freelancers or agencies for urgent resource needs.

External Risks:

- Regularly consult with academic stakeholders to stay updated on potential changes.
- Implement minor changes through the change management process.

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6. Risk Monitoring and Reporting

- **Frequency:** Risks will be reviewed and updated weekly during team meetings.
- **Reporting:** High-priority risks will be escalated immediately to the Project Manager and stakeholders.
- **Metrics:** Track the number of resolved risks versus open risks to ensure effective mitigation.

9.1 RISK LOG

Link to an external risk log or attach a log as an appendix.

10. COST BASELINE

The **Cost Baseline** establishes the approved budget for the project and is used as a reference to monitor and control expenditures throughout the project lifecycle. Below is the detailed cost breakdown for the SIAKAD project:

Cost Breakdown by Project Phase

Project Phase	**Cost Component**	**Estimated Cost (Rp)**	**Comments**
Initiation Phase	Project Manager (Part-time, 1 month)	3,000,000	Includes scope definition, requirements gathering.
	Administrative Costs	500,000	Documentation, meetings, and communication tools.
Design Phase	UI/UX Designer (Freelance)	4,000,000	One-time cost for wireframes and prototypes.
	Design Tools (Figma, Free Plan)	0	Free version sufficient for the project.
Development Phase	Frontend Developer (Full-time, 3 months)	15,000,000	Develop user interface and responsiveness.
	Backend Developer (Full-time, 3 months)	15,000,000	Develop APIs, database, and authentication.
	Tools (Laravel, MySQL, Free Tools)	0	Free-tier tools used during development.
Testing Phase	QA Tester (Part-time, 1 month)	2,000,000	Includes functionality, security, and performance testing.
	Testing Tools (Lighthouse, OWASP ZAP)	0	Free-tier tools for testing purposes.
Deployment Phase	Hosting and Domain (3 months)	575,000	Includes Niagahoster hosting and domain registration.
	Email Notification (SendGrid Free Tier)	0	Free-tier sufficient for notifications.
	Google Analytics	0	Free for tracking system usage.
Miscellaneous Costs	Buffer for Contingencies (5% of total)	2,425,000	Reserved for unforeseen expenses.

Total Estimated Cost

Category	Amount (Rp)
Initiation Phase	3,500,000
Design Phase	4,000,000
Development Phase	30,000,000
Testing Phase	2,000,000
Deployment Phase	575,000
Miscellaneous Costs	2,425,000
Grand Total	**42,500,000**

Budget Monitoring and Control

1. **Tracking Tools:** Use Trello and Excel to track expenditures in real-time against the cost baseline.
2. **Approval Process:** Any costs exceeding the baseline require stakeholder approval.
3. **Reporting Frequency:** Financial status will be reviewed monthly, and a summary will be shared with stakeholders.

11. QUALITY BASELINE

ITEM	ACCEPTABLE LEVEL	COMMENTS
System Performance	Response time under 2 seconds for 90% of transactions.	Measured using Lighthouse and stress testing tools.
Functionality	All modules (e.g., course registration, grade input) must function as specified.	Validated during User Acceptance Testing (UAT).
Usability	System usability score (SUS) of 80 or above.	Feedback collected during UAT with students, faculty, and administrators.

Responsiveness	Fully responsive design across mobile, tablet, and desktop devices.	Tested on Chrome, Firefox, Edge, and Safari browsers.
Data Accuracy	100% accuracy in academic record calculations and storage.	Data integrity tests conducted with sample data during the testing phase.
Security	Zero critical vulnerabilities detected.	Security testing conducted using OWASP ZAP.
Error Rate	Less than 5% bugs in critical functionality post-deployment.	Measured by tracking issues post-deployment via Trello or GitHub issue tracking.
Documentation	Complete and accurate user and administrator manuals.	Reviewed by stakeholders and validated during the training phase.
System Uptime	99.9% uptime after deployment.	Monitored using hosting provider's performance tools.

Baseline Review Process

1. **Stakeholder Review:**
   The quality baseline is reviewed and approved by key stakeholders, including academic administrators and the project sponsor.
2. **Validation Methodology:**
   - Conduct functional testing for modules like course registration and grade entry.
   - Gather usability feedback through UAT sessions.
   - Test responsiveness on multiple devices and browsers.
   - Perform security audits to eliminate vulnerabilities.
3. **Documentation:**
   All test results and stakeholder approvals will be recorded and stored in the project repository.

Quality Monitoring and Reporting

1. **Testing Phases:**
   - Unit Testing: Ensures individual modules meet quality criteria.

    - Integration Testing: Validates the smooth interaction between frontend and backend.
    - User Acceptance Testing (UAT): Confirms the system meets stakeholder requirements.
2. **Tools Used for Monitoring:**
    - Lighthouse: Performance and responsiveness testing.
    - OWASP ZAP: Security and vulnerability scanning.
    - Google Analytics: Tracks system performance post-deployment.
3. **Reporting Frequency:**
    - Weekly progress updates during development and testing phases.
    - Final quality report shared before deployment.

Sign-off Criteria

The project will be considered complete and ready for deployment only when:

- All modules meet the defined functionality and performance benchmarks.
- No critical or high-priority bugs remain unresolved.
- Stakeholders formally approve the final deliverables.

12. APPENDICES

Attach or link to separate plan documents or other reference documents. *Optional.*

ATTACHMENT NAME	LOCATION / LINK

13. AUTHORIZATION SIGNATURES

PREPARED BY

Pramana Yoga Saputra

Name and Title (Printed)

Signature *Date: 19/11/2024*

RECOMMENDED BY

Name and Title (Printed)

Signature *Date*

APPROVED BY

Rosa Andrie Asmara

Project Sponsor Name and Title (Printed)

Project Sponsor Signature *Date: 20/11/2024*